

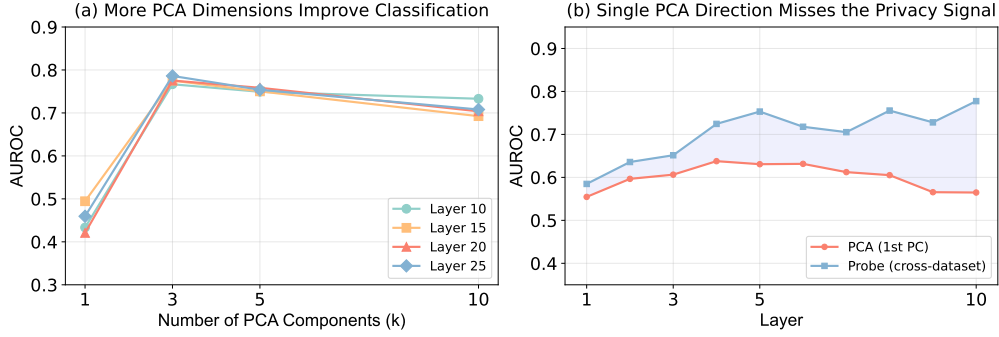


Figure 2: Multi-dimensional privacy on CONFAIDE Tier 2. **Left:** PCA requires $k=3$ components to achieve best results; **Right:** Layer-wise AUROC of probe transfer vs. PCA (1st PC). The probe improves monotonically, while PCA rises only after layer 15.

Why PCA fails. Each concept-level pair varies both recipient and transmission principle simultaneously. PCA on these mixed-parameter differences compresses inherently multi-dimensional variation into a single component, conflating distinct CI parameters. The supervised probe sidesteps this by learning a multi-dimensional decision boundary, which motivates CI-parametric decomposition (§3.4).

3.3 Finding 2 (Behavioral-Level Probing): The Privacy Awareness Gap

Finding 1 shows that contextual privacy norms are linearly separable in the residual stream. We now examine whether this internal signal persists when the model behaves in compliance with CI norms, whether it can generate appropriate responses rather than merely encoding appropriateness judgments. We evaluate Llama-3.1-8B in social role-play scenarios drawn from our behavioral-level stimuli and from CONFAIDE Tier 3. Model outputs are assessed using a GPT-4o-mini judge that labels each response for information leakage, refusal, and contextual integrity compliance. We report NCR (Norm Compliance Rate), defined as the fraction of responses that follow the privacy norm, refusing when sharing is inappropriate, and sharing when it is appropriate.

On 200 balanced behavioral scenarios, Llama-3.1-8B discloses private information in 42.5% of scenarios, yielding an NCR of 57.5%, despite strong evidence from the concept-level probe that the model encodes contextual privacy norms in its internal representations. On CONFAIDE Tier 3, which contains 270 multi-party confidentiality scenarios, the unsteered model leaks in 24.1% of cases.

We term this phenomenon the **privacy awareness gap**. Concept-level probing shows that the model strongly encodes the correct contextual privacy norm in its activations, but this does not reliably translate into norm-compliant behavior. This gap is consistent across models: all four leak in up to 39% of CONFAIDE scenarios despite near-perfect concept-level probe accuracy (Appendix C). This disconnect between representation and action directly motivates intervention via steering.

3.4 Finding 3 (CI-Parametric-Level Probing): Privacy Lives in a CI-Aligned Subspace

Finding 1 shows that contextual privacy is multi-dimensional, and Finding 2 reveals a privacy awareness gap. We now examine whether this multi-dimensional structure aligns with contextual integrity (CI) theory. Using our CI-parametric stimuli, we vary each parameter $p \in \{\text{info.type, recipient, transmission.principle}\}$ in isolation while holding the others fixed, producing per-parameter activation datasets.

We test whether these per-parameter representations occupy distinct subspaces via a *subspace selectivity test* (Appendix G). For each CI parameter, we fit Linear Discriminant Analysis (LDA) on the full hidden states to obtain a four-dimensional discriminant subspace optimized for that parameter’s five categories. We then perform cross-projection: for every

pair of parameters (i, j) , we project parameter i 's stimuli into parameter j 's discriminant subspace and train a five-class classifier using five-fold cross-validation.

A clean diagonal emerges across all four models as shown in Figure 7. Each parameter's subspace achieves near-perfect classification accuracy on its own categories, yet drops to 20% on the other two. Each subspace is therefore selective for its own CI parameter, confirming representational independence. In §5.4, we provide functional validation: steering along individual CI axes increases leakage, whereas combining all three reduces it by 89%.

This finding confirms that the multi-dimensional privacy signal identified in Finding 1 decomposes along CI-parameter axes. Applying PCA to each per-parameter dataset yields per-parameter directions $\mathbf{v}_l^{(p)}$, which serve as the steering axes in §4.2.

4 Steering Contextual Privacy Norms

Our results motivate contextual privacy steering. First, contextual privacy is linearly encoded (§3.2). Second, there exists a privacy awareness gap (§3.3). Third, privacy is represented in a CI-aligned subspace (§3.4). Although the privacy signal is present in the model's representations, it does not reliably translate into behavior. Moreover, its compositional structure suggests the need for parameter-specific intervention.

4.1 Monolithic baselines.

We compare against four strong baselines that steer along a single privacy direction from the linear probe. These include two inference-time methods, additive steering and probe-weighted steering (Zou et al., 2023), as well as two weight-based approaches, LoRRA (Zou et al., 2023) and representation tuning (Ackerman, 2024), which permanently internalize the direction using LoRA adapters. Full implementation details are provided in Appendix D.

4.2 CI-Parametric Steering

Our proposed method exploits the CI-aligned subspace identified in Finding 3 by steering independently along each contextual integrity parameter axis:

$$\mathbf{h}'_l = \mathbf{h}_l + \alpha \sum_{p \in \mathcal{P}} \mathbf{v}_l^{(p)}, \quad (2)$$

where $\mathcal{P} = \{\text{info_type}, \text{recipient}, \text{transmission_principle}\}$, $\mathbf{v}_l^{(p)}$ denotes the unit-normalized per-parameter direction from §3.4, and α controls the steering strength. Steering is applied to the top- k layers ranked by aggregate direction magnitude across all parameters.

CI Parameter Selection CI theory defines five parameters, but they play asymmetric roles: information type, recipient, and transmission principle jointly determine whether a flow is appropriate given the context, while data subject and sender define the actors in a flow without directly governing its normative status. We validate this design choice empirically in Appendix I: extending to all five parameters dilutes the steering signal, reducing in-distribution PPI (privacy protection improvement) from 98.8% to 62.4% and causing catastrophic failure on Llama-2 (leakage increases from 52% to 88%).

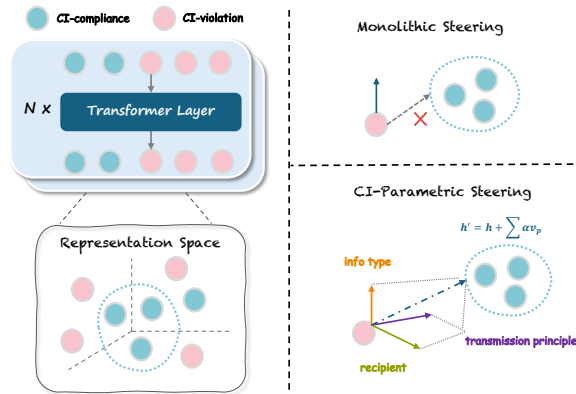


Figure 3: Overview of CI-parametric steering.

As described in Finding 3, this decomposition offers a key advantage over monolithic steering. The dominant source of norm violations can shift across datasets, for example, from recipient-centric violations in synthetic data to transmission-principle-centric violations in CONFAIDE. Per-parameter directions allow the intervention to adapt to these shifts, whereas a single direction fails to transfer. In addition, this decomposition isolates individual CI dimensions, enabling interpretable ablation and analysis (§5.4).

5 Steering Experiments

5.1 Experiment Setup

Datasets. We evaluate on the synthetic dataset used in our probing framework and two contextual privacy benchmarks. The synthetic behavioral dataset contains 200 balanced scenarios. For contextual privacy benchmarks, we use **CONFAIDE (Tier 3)** [Miresghallah et al. \(2024\)](#), which consists of multi-party confidentiality scenarios spanning nine sensitive topics. This benchmark tests whether learned privacy directions transfer to social privacy settings. We also evaluate on **PrivaCI-Bench** [Li et al. \(2025a\)](#), which focuses on prohibited data handling practices under privacy regulations (e.g., GDPR and HIPAA), targeting legal and regulatory scenarios. All steering directions are extracted exclusively from synthetic probing examples and applied to CONFAIDE and PrivaCI-Bench.

LLMs. We evaluate four instruction-tuned models spanning three architecture families: *Llama-3.1-8B-Instruct* [Grattafiori et al. \(2024\)](#), *Qwen-2.5-7B-Instruct* [Yang et al. \(2025\)](#), *Mistral-7B-Instruct-v0.3* [Jiang et al. \(2023\)](#), and *Llama-2-7B-Chat* [Touvron et al. \(2023\)](#). All probing and steering directions are extracted independently per model.

Evaluation. Following [Miresghallah et al. \(2024\)](#), we use GPT-4o-mini (temperature=0) as a judge to classify each model output as *leaked*, *refused*, or *appropriate*, given the full CI context, prompt are listed in [Appendix B](#). We report three metrics.

Leakage Rate ($\downarrow$) = n_{leaked}/N measures the primary safety risk: how often the model discloses private information under a given information flow.

NCR (CI norm compliance rate, $\uparrow$) = $n_{\text{appropriate}}/N$, the fraction of responses deemed contextually appropriate by the judge. NCR is not simply 1 - Leakage Rate, as NCR penalizes all failure modes, including leaking private information, over-refusing when sharing is appropriate, and producing incoherent outputs that are neither informative nor protective. This metric is important because steering methods can reduce leakage by degrading output quality. For example, a model that produces garbled text may achieve low leakage but also low NCR.

PPI (privacy protection improvement) = $1 - \text{Leak}_{\text{steered}}/\text{Leak}_{\text{unsteered}}$ measures the relative reduction in leakage compared to each model’s unsteered baseline, enabling fair comparison across models with different initial leakage rates. PPI > 0 indicates improvement, while PPI < 0 indicates that the intervention degrades privacy.

Hyperparameters. We select the top five layers by probe accuracy as target layers for steering. We sweep $\alpha \in \{0.5, 1.0, 2.0, 4.0\}$ and report results at $\alpha=1.0$ for cross method comparison (Pareto curves in [Appendix F](#)). We use greedy decoding with a maximum generation length of 256 tokens.

5.2 Steering Results on Synthetic Data

[Table 1](#) compares all methods on synthetic data (Llama-3.1-8B, $\alpha=1.0$). Additive steering reduces leakage from 42.5% to 21.5%. Weight-based baselines exhibit inconsistent performance: LoRRA achieves 16.0% leakage with 84.0% NCR, while representation

Table 1: Results on synthetic data.

Method	Leak. (%) $\downarrow$	NCR (%) $\uparrow$	PPI (%) $\uparrow$
No Steering	42.5	57.5	—
Additive	21.5	68.5	49.4
LoRRA	16.0	84.0	62.4
Rep Tuning	37.0	36.0	12.9
CI-Parametric	5.0	90.0	98.8

Table 2: Leakage rate in % ($\downarrow$) on CONFAIDE and PrivaCI-Bench.

		CONFAIDE Tier 3				PrivaCI-Bench			
		Llama-3.1	Qwen-2.5	Mistral	Llama-2	Llama-3.1	Qwen-2.5	Mistral	Llama-2
No Steering		24.1	38.5	25.9	23.7	18.0	16.5	10.7	22.5
LoRRA		31.9	40.0	28.5	15.6	14.0	14.7	16.0	18.7
Rep Tuning		8.5	28.5	3.3	37.4	46.7	18.7	4.0	65.3
$\alpha=0.5$	Add.	47.4	39.6	68.5	58.5	10.0	15.4	40.0	44.0
	CI	2.6	33.0	5.2	27.4	15.3	12.7	2.4	18.7
$\alpha=1.0$	Add.	21.9	39.3	54.6	76.7	64.7	13.3	30.7	73.3
	CI	0.0	15.2	1.9	43.3	0.0	12.0	0.7	19.3

tuning results in 37.0% leakage, worse than inference-time methods. In contrast, CI-parametric steering nearly eliminates leakage (0.5%, PPI = 98.8%). The critical question is whether this advantage persists on CONFAIDE and PrivaCI-Bench.

5.3 Steering Results on CONFAIDE and PrivaCI-Bench

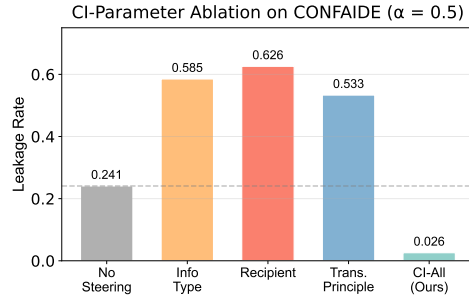
To test whether the CI subspace identified by our probing framework captures generalizable privacy structure rather than memorized template-specific features, we evaluate CI-parametric steering across four models on two contextual privacy benchmarks. Table 2 presents the results.

On CONFAIDE, additive steering *increases* leakage at $\alpha=0.5$ (Llama-3.1: 24.1% $\rightarrow$ 47.4%; Llama-2: 23.7% $\rightarrow$ 58.5%), whereas CI-parametric steering substantially reduces leakage for most models. On PrivaCI-Bench, additive steering on Llama-3.1 *triples* leakage to 64.7% at $\alpha=1.0$, while CI-parametric steering eliminates contextual privacy leakage. These results indicate that a single direction learned from one norm distribution can disrupt partial compliance when transferred to another domain. In contrast, decomposing the privacy signal into per-parameter axes enables targeted control along domain-invariant CI dimensions, leading to more robust generalization.

Weight-based methods such as LoRRA and Rep Tuning further suggest that the bottleneck lies in the steering direction rather than the intervention mechanism. Rep Tuning more than doubles leakage on PrivaCI-Bench for Llama-3.1 (46.7% vs. 18.0%) and Llama-2 (65.3% vs. 16.7%). Permanently internalizing a misaligned direction therefore provides no benefit over inference-time application and can instead amplify privacy violations.

5.4 Analysis

CI-parameter ablation. Figure 4 and Table 4 (in Appendix E) isolate the contribution of each CI dimension. On CONFAIDE Tier 3 ($\alpha=0.5$, Llama-3.1), steering along any *single* CI axis *increases* leakage above the 24.1% baseline. Leakage rises to 58.5% for information type, 62.6% for recipient, and 53.3% for transmission principle, all worse than monolithic steering (47.4%). This mirrors the PCA failure from Finding 1: steering along a single parameter axis disrupts the model’s partially correct multi-dimensional privacy behavior without covering the full CI norm space. The unsteered model already exhibits partial compliance across dimensions; perturbing one axis in isolation pushes the model away from this equilibrium. In contrast, combining all three axes reduces leakage to 2.6% (PPI = 89.2%), confirming that the directions are *complementary*: none is individually sufficient, but together they capture the full privacy subspace.

Figure 4: CI-parameter ablation on CONFAIDE ($\alpha=0.5$).